# Marvel Legends Pricing, 2019–2026: Production Costs, Demand, and the Persistence of Higher Prices

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## Introduction

The broader economic question in this paper is how retail prices in a standardized collector product line relate to production costs, demand conditions, and pricing power. Marvel Legends provides a useful case study because it is a long-running, relatively standardized six-inch line whose standard retail price moved in clear steps rather than changing every month. The standard figure sold for about $19.99 through the first half of 2021, rose to $22.99 in the second half of 2021, moved to $24.99 in 2022, and reached $27.99 in 2025—an increase of roughly 40 percent across the study period. Product announcements and Hasbro management remarks support these analytical price regimes while also showing that older waves sometimes overlapped with newer price tiers (Hasbro, 2021a, 2022a; ToyArk, 2021; ComicBook.com, 2022; MarvelousNews, 2025; see Figure 1).

A simple explanation would be that production became more expensive, so Hasbro raised prices. The data show why that answer is incomplete. Resin, tooling, paint, packaging, freight, labor, tariffs, and exchange rates did not move in the same way around each price increase, while demand conditions also changed. The market was unusually strong during the first two increases, weakened in 2023–2024 without reversing the $24.99 MSRP, and strengthened again around the 2025 increase.

This paper therefore asks: To what extent can Marvel Legends price increases from 2019–2026 be associated with supply-side production costs, and what role did demand conditions play in the timing and persistence of those higher prices? Supply-side measures indicate the pressure manufacturers faced, while demand conditions help describe how much

room the market may have provided to introduce and sustain higher prices. Marvel Legends is the empirical case, not evidence that every action-figure company prices in the same way.

The analysis is intentionally cautious. Public data can describe the economic environment around Marvel Legends, but they cannot recreate Hasbro's confidential cost sheet or full pricing equation. The study does not model licensing and royalties, product development, marketing and selling expenses, retailer economics and margins, promotions, discounts, returns, shared overhead, product mix, or corporate margin objectives because reliable Marvel Legends-level measures are not public. Production-cost proxies and broad demand indicators are therefore partial evidence, not a complete decomposition of MSRP. The goal is to identify which observable pressures aligned most strongly with the actual pricing transitions and to test whether those relationships remain after obvious statistical problems are addressed.

## Methodology

### Sample and price timeline

The primary sample is limited to standard Marvel Legends six-inch single figures sold at the prevailing standard MSRP. Retailer exclusives, deluxe or oversized figures, multipacks, vehicles, bundles, and other nonstandard price tiers are excluded. Build-a-Figure wave figures remain in the sample when the individual figures are sold at the standard price, but the Build-a-Figure parts themselves are treated as embedded wave costs rather than separate predictors.

The monthly study period runs from January 2019 through August 2026. The analytical MSRP regimes are $19.99 through June 2021, $22.99 from July 2021 through March 2022, $24.99 from April 2022 through August 2025, and $27.99 beginning in September 2025. These are treated as retail regimes rather than exact dates on which every retailer and every SKU changed at once (Hasbro, 2021a, 2022a) (see Figure 1).

### Production-cost measures

Monthly public indexes are used as cost-environment proxies rather than as Hasbro invoices. The core measures are the Bureau of Labor Statistics/FRED indexes for thermoplastic resins and plastics materials, tooling and dies, paints, packaging inputs, and deep-sea freight transportation. USD/CNY is retained as exchange-rate context. Packaging is represented by an equal-weight composite of printing ink, adhesives, folding paperboard, and corrugated cartons after each component is normalized to January 2019. Hasbro's 2025 Form 10-K clarifies how these proxies should be interpreted: product costs are generally set through annual pricing agreements with manufacturing partners and typically remain fixed unless significant market changes occur. The indexes therefore measure pressures that can influence negotiated finished-product costs rather than literal spot prices Hasbro necessarily pays for loose resin or other inputs (BLS/FRED, 2019–2026; Hasbro, 2025a).

Labor is treated differently because a credible monthly Marvel Legends labor series does not exist. Manufacturing wage trends from China, Vietnam, and India are used to create a bounded sensitivity range. Hasbro's own cost disclosures provide a corporate-level magnitude check: in 2021 and 2022, respectively, resins were 4.7 and 5.1 percent of total net revenue, labor 5.9 and 5.6 percent, factory overhead 4.1 and 4.8 percent, freight/import costs 3.5 and 4.1 percent, and tooling 1.2 and 1.3 percent. These are shares of total Hasbro net revenue, not Marvel Legends unit-cost weights or shares of MSRP, so they are used only to establish that the categories were economically material (Hasbro, 2022b, 2023; National Bureau of Statistics of China, 2021–2026; National Statistics Office of Vietnam, 2026; MOSPI, 2025).

Tariffs are kept separate from freight. Freight measures the cost of moving goods; tariffs are government-imposed import costs. The tariff analysis uses the timing of China-related policy

changes together with Hasbro's reported tariff expense. Hasbro recognized approximately $44.9 million of tariff costs in 2025, which provides a real company-level magnitude anchor without implying a direct dollar amount for one Marvel Legends figure (Hasbro, 2025a, 2025b).

### Demand measures

Demand is harder to observe because there is no public monthly Marvel Legends unit-sales series for the full period. The study therefore combines several imperfect measures: annual U.S. Action Figures & Accessories retail sales, available toy-industry unit growth and average selling price information, Hasbro action-figure point-of-sale and category-share evidence, and a monthly U.S. Hobby, Toy and Game Store retail-sales series transformed to year-over-year growth. Dollar sales are not treated as pure demand because higher prices themselves can raise sales dollars (Circana/The Toy Association, 2023–2026; U.S. Census Bureau/FRED, 2018–2026).

### Statistical approach

The analysis uses both regression and transition windows because the dataset contains 92 monthly observations but only three upward MSRP decisions. The economic target can be summarized as:

$$(1) \quad P_t = f(C_t, D_t, Z_t)$$

Here, P is standard MSRP at time t, C is observable production-cost pressure, D is demand conditions, and Z is unobserved or unavailable line-level factors. The study does not estimate missing dollar values for C or Z. Public proxies instead measure the direction, relative magnitude, and timing of pressure around known MSRP changes; the analysis classifies the impact environment rather than reconstructing Hasbro's confidential cost sheet or profit statement.

Production-cost pressure is expected to be positively associated with MSRP changes, although timing and strength can differ by input. Demand is expected to matter mainly for the market's ability to absorb and sustain a higher price tier.

The preferred specification uses twelve-month percentage changes in MSRP, resin, tooling, and freight, with year-over-year toy/hobby retail sales as a broad demand control:

$$(2)\quad \Delta_{12}\,MSRP_t = \alpha + \beta_1\,\Delta_{12}\,Resin_t + \beta_2\,\Delta_{12}\,Tooling_t + \beta_3\,\Delta_{12}\,Freight_t + \beta_4\,Demand_t + \varepsilon_t$$

$$\text{where}\quad \Delta_{12}\,X_t = (X_t / X_{t-12}) - 1.$$

A raw-level regression is retained only as a diagnostic because MSRP and many cost indexes trend upward together. Annual differencing reduces that common-trend problem, and HAC/Newey-West standard errors with twelve lags address serial dependence from overlapping changes. Common supply lags of 0, 3, 6, 9, and 12 months and six- and twelve-month windows before the three price transitions test timing without choosing separate "best" lags. This strategy estimates conditional associations, not causal effects.

Endogeneity remains an important limitation. Some regressors may be jointly determined with price or move with unobserved factors. The demand proxy uses retail dollar sales, which can rise partly because prices rise, creating potential reverse causality. Omitted factors such as licensing, retailer margins, product mix, and pricing objectives may affect both MSRP and observed market conditions. The coefficients are therefore interpreted as conditional associations rather than causal estimates.

**How We Handled Missing Product-Level Data**

Hasbro does not publicly disclose the exact factory cost of a Marvel Legends figure, the price of each mold, the labor time per unit, or the plastic recipe. Rather than fill those gaps with

guesses, the study uses the closest defensible public evidence and states what each proxy can and cannot establish.

**How do we estimate mold and tooling costs if Hasbro does not tell us what a mold costs?**

The study does not guess a Marvel Legends mold price. An industry tooling-and-dies index tracks the broader tooling-cost environment, while Hasbro's 2025 Form 10-K confirms that Hasbro generally owns product tools, dies, and molds and depreciates them over useful lives generally around three years. That establishes tooling as a real capital cost without turning a company-wide asset balance into a Marvel Legends invoice. Mold reuse evidence is therefore qualitative only (BLS/FRED, 2019–2026; Hasbro, 2025a; Vala/Valaverse, 2024).

**How do we handle labor when we do not know what Marvel Legends factory workers were paid?**

Because Marvel Legends factory payroll is not public, the study uses manufacturing wage changes in China, Vietnam, and India as a bounded sensitivity range rather than constructing a false monthly payroll series. The range acknowledges uncertainty in country mix, labor time per figure, benefits, yield, and contract rates.

**Can we use Hasbro's own cost information?**

Hasbro's historical disclosures confirm that labor, resin, freight/import costs, tooling, and factory overhead were meaningful company-level production categories. They are used to judge economic importance, not to allocate a percentage of a $24.99 or $27.99 figure to any one input.

**What about Build-a-Figure parts, accessories, plastics, petroleum, and packaging?**

Accessories and Build-a-Figure parts are treated as embedded costs because their actual unit costs are not public and BAF costs are spread across a wave. A broad thermoplastic-resin index is used instead of guessing the PVC/ABS/POM mixture. Marvel Legends designer Dwight Stall's 2025 explanation that multipart ABS construction can reduce weight while requiring additional tooling illustrates why less material does not necessarily mean lower total production cost. Petroleum is treated as an upstream driver of plastics and freight rather than a duplicate core variable; packaging inputs are combined because their indexes move closely together; and windowless packaging remains background context because no reliable savings-per-figure estimate is public (Reif, 2025).

**How do we measure demand if we do not have Marvel Legends sales numbers?**

No public series is treated as "the Marvel Legends demand number." Action-figure sales, unit and average-selling-price evidence, Hasbro point-of-sale information, and broader toy/hobby retail sales are read together to judge whether the market was strengthening or weakening without equating category dollars with Marvel Legends unit demand.

The governing rule is simple: when the exact number is not public, use a transparent proxy or range and do not manufacture precision.

**Results**

Before comparing the three price increases, it is useful to see the scale and timing of the main cost movements. Figure 2 plots the principal production-cost proxies on a common 2019-average=100 basis, while Exhibits 1–3 summarize study-period changes, Hasbro accounting context, and the twelve-month windows before each MSRP transition. These are market indexes and company-level benchmarks, not Marvel Legends factory invoices.

**Exhibit 1. Study-period changes in major price and cost measures**

| Measure | 2019–2026 change |
|---|---|
| Marvel Legends standard MSRP | +40.0% |
| Thermoplastic plastics / resin PPI | +19.0% |
| Tooling / dies PPI | +33.0% |
| Paints PPI | +49.1% |
| Printing ink PPI | +28.8% |
| Adhesives PPI | +49.4% |
| Paperboard PPI | +42.8% |
| Corrugated cartons PPI | +42.5% |
| Packaging composite | about +41% |
| Deep-sea freight PPI | +68.8% |
| USD/CNY | about -0.7% |

*Note. PPI and exchange-rate series are public market proxies. Percentage changes describe the study window and should not be read as Hasbro-specific pass-through rates. Sources: BLS/FRED, Census/FRED, and the study workbook.*

**Exhibit 2. Selected Hasbro production-cost categories as a share of cost of goods sold**

| Cost category | 2019 | 2020 | 2021 | 2022 |
|---|---|---|---|---|
| Labor | 21.1% | 19.7% | 19.7% | 17.2% |
| Resin | 13.3% | 12.4% | 15.7% | 15.6% |
| Freight / import | 6.8% | 7.3% | 11.7% | 12.6% |
| Tooling | 4.7% | 4.1% | 4.0% | 4.0% |

*Note. These are derived from Hasbro company-wide cost-of-sales disclosures, not Marvel Legends-specific cost shares. They are used only to show that labor, resin, freight/import, and tooling were economically meaningful production categories. Source: Hasbro cost-of-sales disclosures, 2019–2022.*

**Exhibit 3. Cost changes in the 12 months before each Marvel Legends MSRP increase**

| Cost measure | Jul. 2021 $19.99 → $22.99 | Apr. 2022 $22.99 → $24.99 | Sep. 2025 $24.99 → $27.99 |
|---|---|---|---|
| Resin | +57.7% | +3.6% | -1.0% |
| Tooling | +3.0% | +9.4% | +4.8% |
| Paint | +6.1% | +18.3% | +3.6% |
| Packaging | +5.1% | +14.5% | +3.1% |
| Freight | +16.7% | +15.2% | -5.2% |

*Note. Values are 12-month changes in the public cost proxies immediately preceding each analytical MSRP breakpoint. The contrast is central to the paper: 2021 is dominated by resin and freight, 2022 by broader production inflation, and 2025 by a much weaker resin/freight backdrop.*

### 2021: the clearest material and freight shock

The first major price increase, from $19.99 to $22.99 in July 2021, has the strongest traditional production-cost backdrop in the study. During the twelve months before the transition, the resin index rose about 58 percent and deep-sea freight about 17 percent. Tooling, paint, and packaging increased much less. Hasbro also stated that higher input and freight costs were driving price increases and that those increases were being implemented during the third quarter of 2021 (Hasbro, 2021a).

Demand was also unusually strong. U.S. Action Figures & Accessories retail sales rose from roughly $1.8 billion in 2020 to about $2.3 billion in 2021. That does not prove that demand caused the price increase, but it shows that the increase occurred in a strong category environment rather than a weak one (Circana/The Toy Association, 2026a).

### 2022: broader production inflation

The April 2022 increase from $22.99 to $24.99 looks different. Over the prior twelve months resin rose only about 4 percent, while tooling increased roughly 9 percent, paint 18 percent, packaging 15 percent, and freight 15 percent. The second increase therefore followed broader production inflation rather than another resin spike. Hasbro had already announced

additional price increases for the second quarter of 2022, and standard $24.99 figures were appearing in new product solicitations (Hasbro, 2022a; ComicBook.com, 2022).

Action-figure dollar sales reached about $2.4 billion in 2022, the high point of the study-period annual series. Broader toy-market evidence also shows why dollar sales need care: total toy dollars were roughly flat in 2022 while units fell about 4 percent and average selling prices rose about 3 percent (Circana/The Toy Association, 2023).

**2025: a different mix of pressures**

The September 2025 move from $24.99 to $27.99 is the break in the earlier pattern. In the preceding twelve months resin was roughly flat and freight was down about 5 percent. Tooling, paint, and packaging rose only about 3 to 5 percent. The earlier resin-and-freight story therefore does not explain the 2025 increase nearly as well.

Other pressures were more important by then. Hasbro recognized about $44.9 million in tariff costs during 2025 and had discussed targeted pricing as one possible mitigation tool before the September increase. Labor benchmarks also remained well above their 2019 levels. At the same time, demand was improving: 2025 U.S. toy dollars rose about 6 percent, units about 3 percent, and average selling prices about 4 percent (Hasbro, 2025a, 2025b; Circana/The Toy Association, 2026a).

Labor also remained economically meaningful even though it could not be measured as a monthly Marvel Legends series. The study's bounded sensitivity analysis implies roughly 4.0% to 8.1% potential labor-driven COGS pressure over the period, with about 6.0% as the middle scenario. These are sensitivity anchors, not product-line cost estimates.

**Regression and robustness results**

**Exhibit 4. Preferred annual-change regression**

| Predictor | Standardized beta | HAC p-value |
|---|---|---|
| Resin | +0.173 | .091 |
| Tooling | +0.315 | .041 |
| Freight | +0.478 | < .001 |
| Demand proxy | -0.076 | .143 |

*Model summary: N = 79; $R^2$ = .630; adjusted $R^2$ = .610; maximum VIF = 2.29. HAC/Newey-West standard errors use 12 lags. The rejected raw-level model produced $R^2 \approx .959$ but a maximum VIF ≈ 185, which is why it is treated as a warning rather than the headline result.*

The raw-level regression produced an $R^2$ of about 0.96, but that result was rejected as a headline finding. The maximum variance inflation factor was about 185, and closely related predictors such as paint and packaging received unstable coefficients. The model mostly captured the fact that MSRP and many inputs rose over time.

The preferred annual-change model is more credible. It produced an $R^2$ of about 0.63 and an adjusted $R^2$ of about 0.61, with a maximum VIF of about 2.29. Freight had the strongest standardized same-period association with MSRP inflation (about 0.48), followed by tooling (about 0.32). Resin was positive but weaker in the same-period model (about 0.17). The broad demand control was small and statistically unreliable. Removing the demand control did not materially change the main supply-side pattern.

Lag tests add a timing result rather than a new causal claim. Freight and tooling were strongest at current or short common lags, while resin became more strongly positive at longer common lags. A mixed-timing model did not cleanly confirm one exact resin delay, so the study does not select a separate “best” lag for each cost. The regression is therefore supporting evidence, not a formula for converting costs into dollars of MSRP.

### Demand Conditions and the Persistence of Higher Prices

Demand adds a different part of the explanation. Figure 3 places the monthly toy/hobby retail-sales proxy against the Marvel Legends price regimes. Higher costs can create pressure to raise price, but a company still needs customers willing to buy at the new price. The first two increases occurred while action-figure spending was unusually strong, helping explain how higher price tiers could be absorbed by the market.

The more interesting period is 2023–2024. Action-figure dollar sales fell from roughly $2.4 billion in 2022 to $2.2 billion in 2023 and $2.0 billion in 2024, while broader toy-industry reports described weaker discretionary spending and then stabilization. Marvel Legends MSRP nevertheless remained at $24.99 (Circana/The Toy Association, 2024, 2025). This is not formal proof of downward price rigidity, because retailer discounts, product mix, licensing, and internal pricing strategy are not fully observed. It does show that weaker category conditions did not reverse the established standard price.

Demand strengthened again by 2025. Using the rounded annual figures in this study, Action Figures & Accessories sales rose from about $2.0 billion to $2.1 billion. More importantly, total toy-industry units returned to growth. Hasbro remained the leading action-figure supplier in its cited Circana markets, and Q1 2026 Action Figures & Accessories point-of-sale growth was about 12.8 percent, with Marvel listed among growth brands. Through April 2026, U.S. toy dollars were up about 13 percent, units about 5 percent, and adults 18+ accounted for 35 percent of industry growth (Hasbro, 2026; Circana/The Toy Association, 2026b).

The monthly demand control does not perform strongly in the regression, but that should not be read as evidence that demand did not matter. It is a broad retail-dollar proxy covering

much more than Marvel Legends. Demand is more useful here as context for the market surrounding each pricing decision than as one coefficient in the monthly model.

### Discussion and Interpretation

The main finding is not simply that production costs rose. The economic explanation changes across the three price increases, which is also visible in the regime averages in Figure 4. The 2021 increase followed the clearest resin and freight shock; the 2022 increase followed broader production inflation; and the 2025 increase occurred after the earlier resin/freight pressure had weakened, while tariffs, accumulated labor pressure, and improving demand had become more relevant.

This difference matters because it argues against one stable pass-through story. If one production cost were the main explanation for the entire 40 percent MSRP increase, a similar cost pattern should appear around each transition. It does not. The supply-side explanation is strongest in 2021–2022 and more mixed in 2025.

The period after 2022 also raises a second point. Some extraordinary pandemic-era cost pressures eased, and action-figure sales weakened, but the standard $24.99 MSRP did not fall. The available data cannot prove why Hasbro maintained that price. Still, the pattern suggests an asymmetry worth noting: rising costs can help establish a higher price tier, while falling costs do not necessarily bring the old price back.

The statistical work supports this interpretation but does not replace it. Rejecting the 96 percent raw-level fit was important because a model can look excellent simply because everything trends upward together. The annual-change model leaves a meaningful supply relationship in place, especially for freight and tooling, while the transition windows show that the mix of relevant pressures changed from one pricing decision to the next.

Demand completes the picture. Strong demand does not prove that Hasbro raised prices because collectors would tolerate them. It does, however, affect how easily a higher price can be introduced and sustained. The evidence therefore supports a supply-and-demand interpretation rather than a production-cost-only explanation: supply pressure helps explain why prices moved upward, while demand helps explain the market environment in which those new price levels persisted.

### Practical Use and Recommendation

The framework is useful beyond evaluating Hasbro's past pricing decisions. Public market data cannot replace a factory quote, but they can help investors, project managers, and smaller manufacturers judge whether quoted tooling, materials, labor, packaging, and freight costs are broadly consistent with the current production environment.

The scale difference matters. A smaller action-figure company may produce far fewer units than Hasbro and therefore face higher unit costs or weaker tooling amortization. The results should not be read as proof that another firm could reproduce Marvel Legends at a lower price and automatically earn a profit. Production cost, expected volume, acceptable retail price, and required operating margin must be evaluated together.

For managers and investors, the main recommendation is not to assume that every retail-price increase reflects the same cost pressure. The Marvel Legends transitions occurred under different combinations of resin, freight, tooling, broader inflation, tariffs, labor pressure, and demand. A useful pre-production process is therefore to benchmark major inputs, compare sourcing alternatives, test scale and price scenarios, and use outside data to ask better questions of manufacturer quotes.

This research cannot determine the ideal price of a future product or substitute for confidential accounting. It can provide a more informed starting point for separating production-cost pressure from the demand and pricing-power conditions that determine whether a higher retail price can be sustained.

## Conclusion

Marvel Legends pricing from 2019 through 2026 cannot be explained by one production-cost variable or a stable one-for-one cost-pass-through rule. The clearest result is that the three MSRP increases occurred under materially different cost environments, and the higher price tiers persisted even after some of the pressures associated with earlier increases receded. Production costs mattered, especially in 2021 and 2022, but their importance and timing changed across the three transitions.

The 2021 move to $22.99 followed the clearest resin and freight shock and was directly consistent with Hasbro's public comments about higher input and shipping costs. The 2022 move to $24.99 followed broader inflation across tooling, paint, packaging, freight, and labor. The 2025 move to $27.99 occurred under a different mix: resin was roughly flat and freight was lower, while tariffs, accumulated labor pressure, and stronger demand were more relevant.

The preferred regression supports a partial and time-varying supply relationship rather than exact cost pass-through. Demand is harder to measure and its monthly regression coefficient is weak, so the study does not assign demand a numerical share of any MSRP increase. The broader evidence nevertheless shows that demand conditions were relevant to market absorption and persistence: the first two increases occurred during strong action-figure spending, weaker demand in 2023–2024 did not reverse the $24.99 MSRP, and demand strengthened again around the $27.99 regime.

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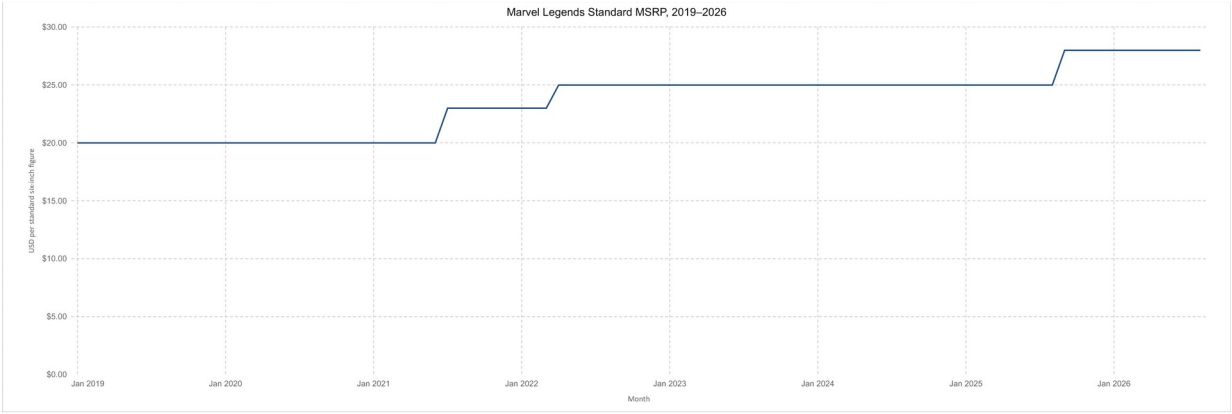
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## Figures

**Figure 1. Marvel Legends Standard MSRP, 2019–2026.**

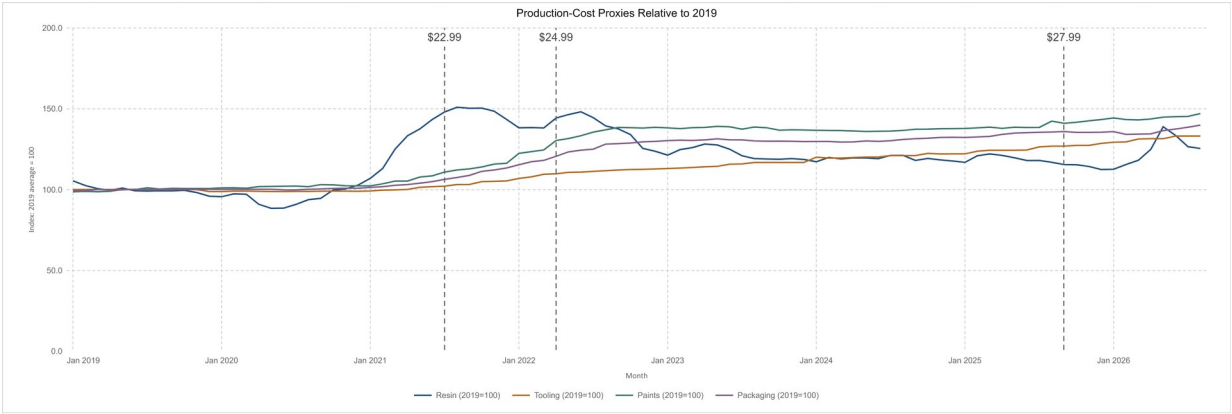



Standard MSRP; four analytical price regimes.

Regimes: $19.99 Jan 2019–Jun 2021; $22.99 Jul 2021–Mar 2022; $24.99 Apr 2022–Aug 2025; $27.99 Sep 2025–Aug 2026.

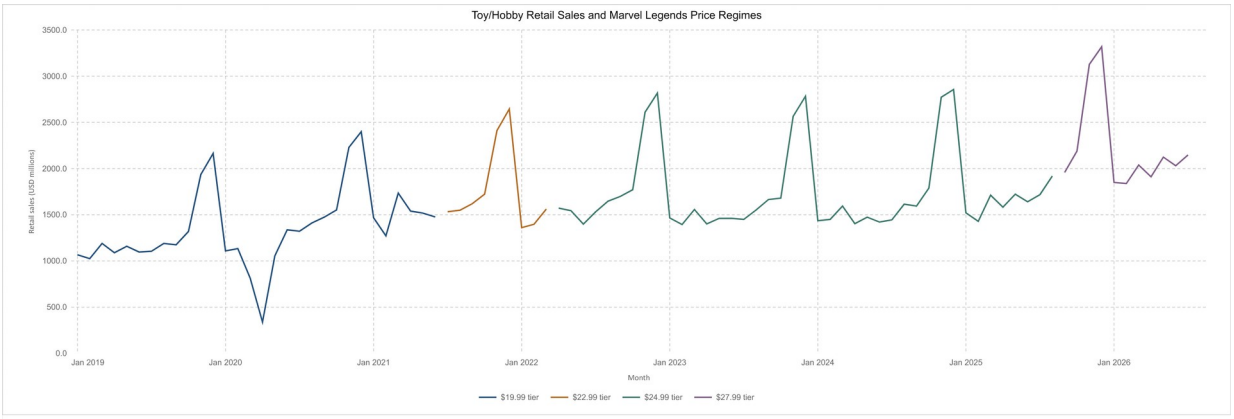
Note. Analytical MSRP regimes are used to summarize the standard six-inch line; individual retailer and SKU transitions can overlap. Source: study workbook and MSRP timing sources listed in the References.

**Figure 2. Production-Cost Proxies Relative to 2019.**



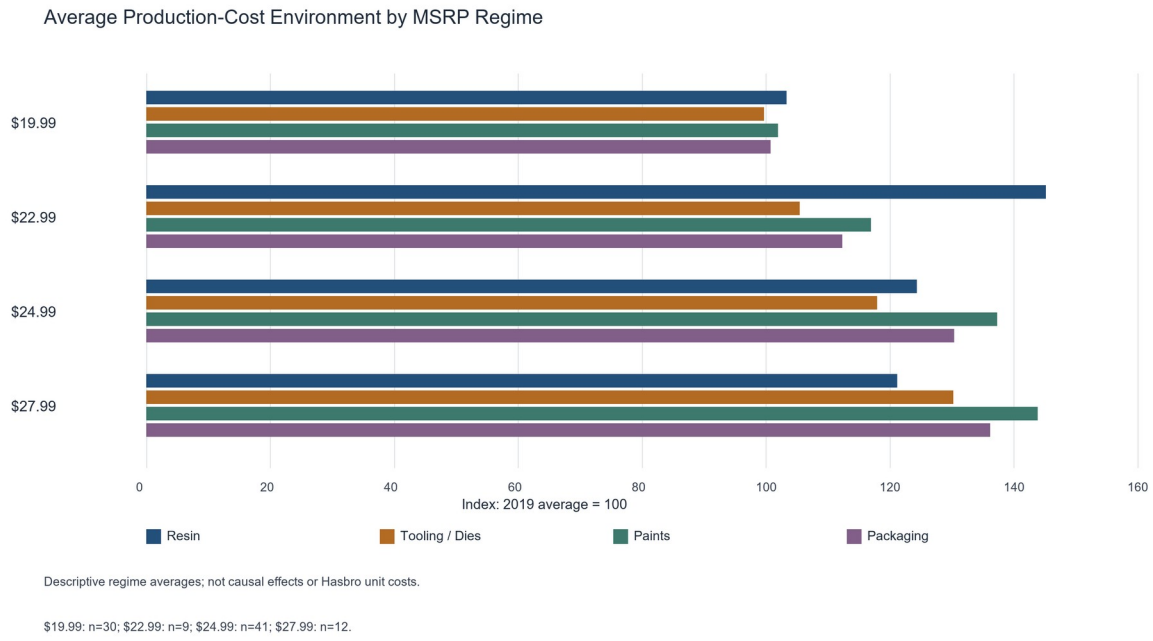

Note. Monthly public indexes are normalized to a 2019 average of 100. They describe the production-cost environment and are not Hasbro or Marvel Legends unit-cost indexes. Dashed lines mark the three analytical MSRP transitions.

**Figure 3. Toy/Hobby Retail Sales and Marvel Legends Price Regimes.**




Note. The retail-sales series is a broad demand-side contextual measure, not Marvel Legends sales. Source: U.S. Census Bureau/FRED and the study workbook.

**Figure 4. Average Production-Cost Environment by MSRP Regime.**




Note. Values are descriptive averages of normalized production-cost proxies within each MSRP regime. They are not causal effects or Hasbro unit-cost shares. Regime observations: $19.99 n=30; $22.99 n=9; $24.99 n=41; $27.99 n=12.

## Appendix A: Research Decisions and Statistical Checks

This appendix explains the main statistical choices in plain English. Each technical term is followed by the practical question it was meant to answer. The frozen V20 workbook retains the full source register, series IDs, date coverage, transformations, and source URLs used to construct the quantitative analysis; the final chart workbook retains the regime and annual pivots, correlation diagnostics, and chart notes.

### 1. Multicollinearity diagnostics

*Are my predictors too similar to each other?*

Several production-cost variables rose together. Variance Inflation Factors (VIFs) were used to check whether the regression could reliably separate them. The raw-level model had a maximum VIF of about 185, which is severe. The preferred annual-change model reduced the maximum VIF to about 2.29, a much more manageable level.

### 2. Why the 96 percent regression was rejected

*Does a model that fits almost perfectly necessarily tell me something useful?*

No. The raw-level model produced an $R^2$ near 0.96 largely because MSRP and many input indexes trended upward over the same years. The model was kept as a warning example, not as headline evidence. A great fit is not useful if it mostly says everything went up together.

### 3. Annual changes

*What happens if I compare changes instead of just comparing rising levels?*

The preferred model uses twelve-month percentage changes. That asks whether changes in cost pressure line up with changes in MSRP inflation. Adjusted $R^2$ falls to about 0.61, but the result is more credible because it depends less on the common upward trend.

### 4. Lag tests

*Does the past influence the present in my time-series data?*

Supply variables were tested at common lags of 0, 3, 6, 9, and 12 months. Freight and tooling were strongest at current or short lags, while resin became more positive at longer lags. This suggests different costs may reach pricing at different stages, but the study does not claim one exact causal delay.

### 5. No lag cherry-picking

*Did we simply choose whichever delay made each variable look best?*

No. Common lag structures were compared as a family, and the zero-lag annual-change model remained the baseline. This avoids choosing a different lag for each variable simply because it produces the most attractive result.

### 6. HAC / Newey-West standard errors

*Am I correcting for the fact that nearby months are related to each other?*

Yes. Twelve-month changes overlap from month to month, so observations are not fully independent. HAC/Newey-West standard errors with twelve lags provide a more cautious measure of uncertainty under that dependence.

### 7. Transition-window analysis

*What was happening before each actual price increase?*

The dataset has 92 monthly observations but only three upward price decisions. Six- and twelve-month windows before July 2021, April 2022, and September 2025 are therefore used alongside regression results so that 92 rows are not mistaken for 92 separate Hasbro pricing choices.

### 8. Packaging composite

*Why not put ink, adhesives, paperboard, and cartons in the regression separately?*

The four packaging series move extremely closely together. Combining their normalized values into one packaging measure makes the model easier to interpret and avoids forcing nearly identical predictors to compete for the same statistical credit.

### 9. Labor sensitivity analysis

*How do we include labor when Marvel Legends factory payroll is not public?*

National manufacturing wage trends and historical Hasbro labor shares are used to create low, base, and high sensitivity ranges. Labor stays in the economic story without pretending that a monthly Marvel Legends payroll series exists.

### 10. Petroleum and double counting

*Am I accidentally counting the same cost pressure twice?*

Petroleum can affect both plastics and freight. Because resin and freight are already measured directly, petroleum is used as an upstream explanation rather than another core predictor.

### 11. Demand control

*Could the apparent supply relationship actually be capturing demand?*

The preferred regression includes year-over-year hobby/toy/game retail sales as a broad demand control. Removing it leaves the main supply pattern broadly intact. This does not prove demand is unimportant; it shows the supply result is not being created by that one noisy demand proxy.

### 12. Robustness checks

*Do the conclusions survive reasonable changes in the analysis?*

The study tested alternate common lags, a packaging alternative, a broad-cost composite, a supply-only model, and a sample ending before the September 2025 regime. The purpose was to see whether the interpretation changed, not to hunt for the most significant model.

### 13. Association versus causation

*Does the regression prove that a particular cost caused Hasbro to raise MSRP?*

No. Costs, demand, inflation, tariffs, expectations, and pricing decisions can move together. The paper therefore uses language such as associated with, aligned with, and consistent with rather than treating regression coefficients as causal pass-through rates.

### 14. Reproducibility validation

*Can someone else run the same analysis and get the same numbers?*

Yes. The frozen V20 workbook and fixed specifications were independently rerun through a separate Python validation pipeline. All five supplied tests passed, and the preferred coefficients, HAC standard errors, VIFs, lag models, and robustness results reproduced with no material numerical differences. The validation confirms internal reproducibility; it does not independently certify every external source or turn the associations into causal findings.

### 15. Why these decisions matter

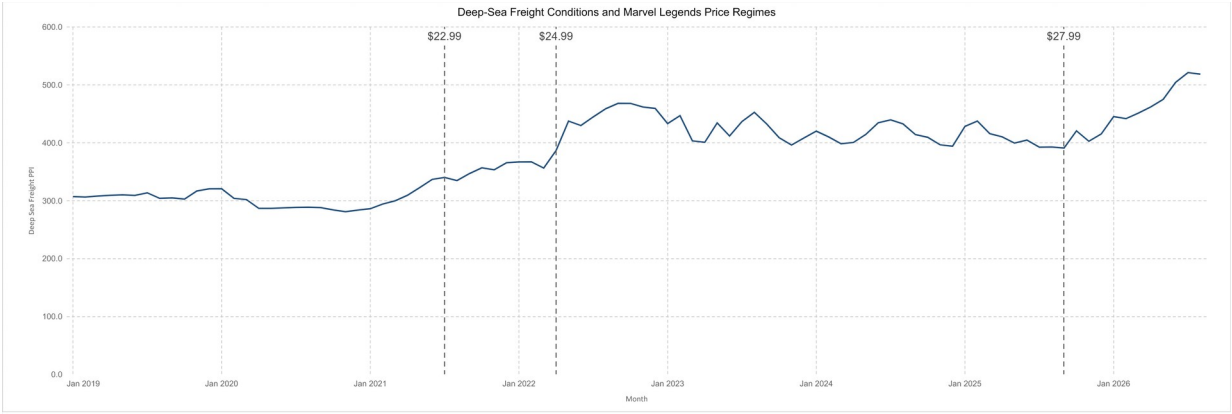
The purpose of these choices was not to make the results look stronger. In several cases, the opposite happened. The 96 percent regression was rejected even though it looked impressive. Labor was reported as a range rather than a single number. Demand stayed in the discussion even though its broad monthly coefficient was weak. Longer resin lags were treated cautiously rather

than promoted as a new headline model. The rule throughout was simple: when the available evidence cannot support precision, the analysis should not manufacture it.

The independent validation report concluded that the V20 quantitative results were reproducible exploratory associations, while emphasizing that only three upward price decisions exist and that the analysis does not establish causal identification (Independent Validation Report, October 2, 2026).

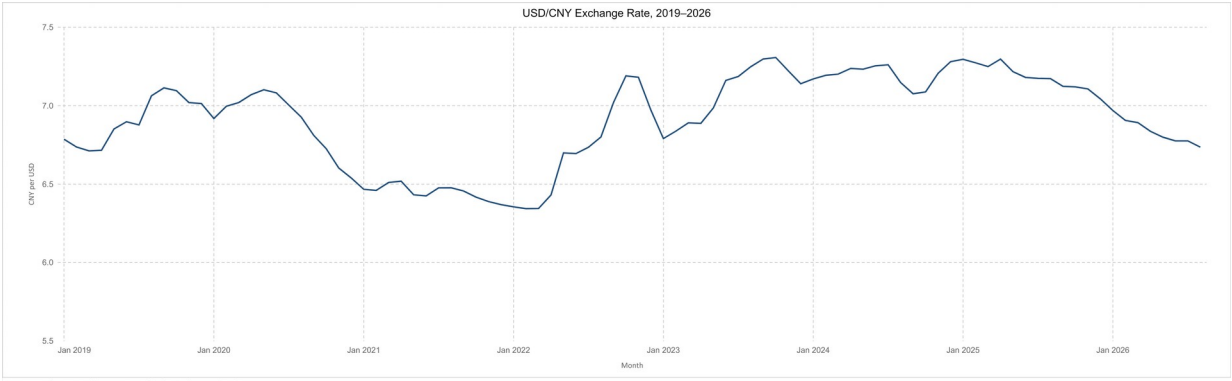
## Appendix B: Supporting Figures

**Figure B1. Deep-Sea Freight Conditions and Marvel Legends Price Regimes.**



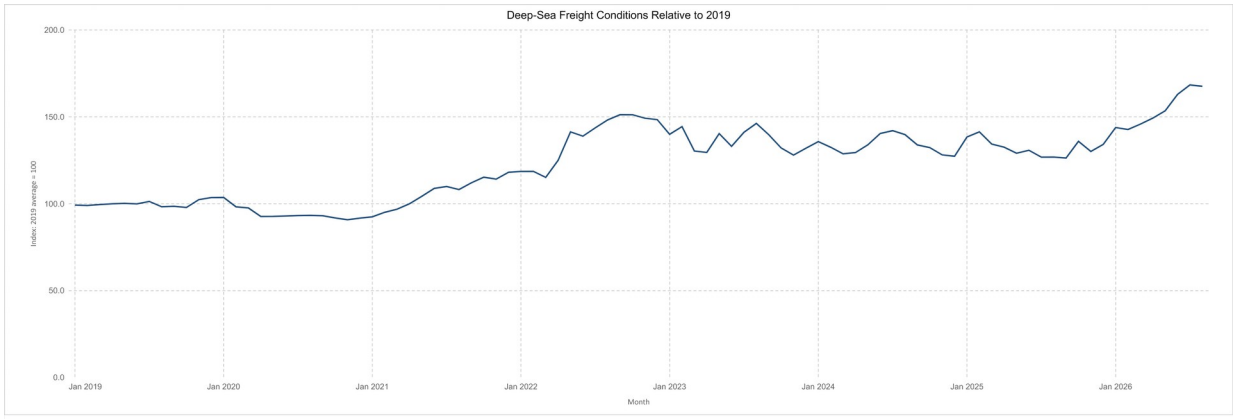

Note. Deep-sea freight is analyzed separately because pandemic-era volatility can dominate the scale of other cost series.

**Figure B2. USD/CNY Exchange Rate, 2019–2026.**




Note. Exchange rates are sensitivity/context evidence and do not establish Hasbro's realized currency exposure.

**Figure B3. Normalized Deep-Sea Freight Conditions.**




Note. Freight is normalized for comparability with other indexed series; this figure is supplementary to Figure B1.

## Appendix C: Data Provenance Summary

This summary identifies the main data families used in the paper. Exact URLs, coverage dates, transformations, and access notes are preserved in the Source Register sheet of the frozen V20 workbook.

| Measure / evidence | Primary source / identifier | Use in study |
|---|---|---|
| Marvel Legends MSRP regimes | Hasbro management remarks; ToyArk; ComicBook.com; MarvelousNews | Defines $19.99, $22.99, $24.99, and $27.99 analytical retail regimes. |
| Thermoplastic resin / plastics | BLS/FRED WPU0662 | Monthly production-cost proxy; annual changes and transition windows. |
| Tooling / dies | BLS/FRED PCU3335143335140 | Monthly tooling-cost proxy; annual changes and transition windows. |
| Paints | BLS/FRED WPU062 | Monthly production-cost context and normalized figures. |
| Packaging composite | BLS/FRED WPU06790919, WPU067904, WPU09150336, WPU09150301 | Equal-weight normalized composite of ink, adhesives, paperboard, and corrugated cartons. |
| Deep-sea freight | BLS/FRED PCU483111483111 | Monthly logistics proxy; regression, transition windows, and figures. |
| Toy/hobby retail sales | U.S. Census Bureau/FRED MRTSSM45112USN | Broad monthly demand control and contextual figure; not Marvel Legends sales. |
| Manufacturing labor | China NBS; Vietnam NSO; India MOSPI/ASI | Annual country wage trends used for bounded sensitivity, not a monthly Legends payroll series. |
| Hasbro production-cost context | Hasbro annual/investor cost-of-sales disclosures, 2019–2022; 2025 Form 10-K | Corporate magnitude checks; supplier-pricing and tooling ownership/depreciation context. |
| Tariffs | Hasbro 2025 Form 10-K and 2025 management remarks; policy timeline logged in V20 | Company-level tariff-cost anchor and 2025 timing context. |
| Engineering mechanism | Reif (2025) interview with Dwight Stall | Legends-specific PVC/ABS, material-weight, parts, and tooling tradeoff. |
| Reproducibility | Independent Validation Report (Oct. 2, 2026) | Independent rerun of fixed V20 specifications; confirms internal numerical reproducibility, not causal identification. |